
Effects of Facial Alignment on Facial Attribute Classification

Rufus Walsh *The University of Sydney, Sydney, Australia*

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1 Abstract

Facial alignment is a common preprocessing technique in computer vision, designed to normalise facial images into a canonical pose to improve recognition accuracy. This study quantitatively analyses the impact of Dlib-based facial alignment on the performance of three common Convolutional Neural Network (CNN) architectures: ResNet50, VGG16, and EfficientNet-B0. The models were tasked with facial attribute classification on the CelebA dataset. We compared the performance (Mean Square Error and accuracy) of models trained on raw images versus images aligned with varying padding levels (0.25, 0.5, and 0.75). The results indicate that facial alignment had a minimal overall impact on performance for the relatively clean CelebA dataset, with the best-aligned models showing an average accuracy of 86.6%-87.7% compared to 86.7%-87.8% for raw images. However, a strong correlation was found between the padding parameter and model accuracy. Newer architectures (ResNet50, EfficientNet-B0) benefited from more focalised crops (higher padding), while the VGG16 model preferred less. This suggests that while alignment may not be critical for high-quality datasets, model architecture plays a significant role in how focalised facial features should be presented.

2 Introduction

Recent advancements in automated facial recognition, driven by the success of applying Convolutional Neural Networks (CNNs) to image classification, have spurred applications in modern security and human-computer interaction. A persistent challenge is bridging the gap between clean experimental samples and variable, "in-the-wild" images.

Preprocessing techniques are often employed to man-

age this variability. Facial alignment is one such technique, which aims to normalise facial images by transforming key features (like eyes, nose, and mouth) into a canonical pose. This reduces pose and scale variance, theoretically lowering the demand on the classification model to learn invariance and allowing it to focus on identity-specific features.

This study has two primary aims:

- To quantitatively analyse the performance of common CNN models (ResNet50, VGG16, and EfficientNet-B0) on a facial attribute classification task, comparing performance with and without facial alignment preprocessing.
- To determine what level of facial alignment "padding" (i.e., the amount of context around the cropped face) provides the optimal performance boost.

3 Methods

3.1 Facial Alignment

Facial alignment is the process of transforming a facial image so that key facial landmarks are registered to fixed locations in a 'canonical' coordinate system. By normalising pose and scale, alignment allows the neural network to learn identity- or expression-specific features more effectively, rather than dedicating model capacity to learning pose invariance.

This study utilised the Dlib library King, 2009 to perform alignment. The process involved:

1. Detecting all faces in an image using Dlib's default face detector.
2. Selecting the largest detected face (if multiple were present).
3. Applying Dlib's 68-point shape predictor Kazemi and Sullivan, 2014 to identify facial landmarks.



(a) *Raw Image*



(b) *Face Aligned with 0.25 padding*



(c) *Face Aligned with 0.5 padding*



(d) *Face Aligned with 0.75 padding*

Figure 1: *Resulting images from varying the amount of padding.*

4. Using Dlib's `get_face_chip` function to perform the necessary transformations (rotation, scaling, and translation) to warp these landmarks to a predefined template.

The padding parameter was a key variable, controlling the amount of contextual space around the aligned face. In this study's implementation, the padding value scaled the crop size such that a padding of 0.5 doubled the crop's width and height relative to the detected face, and a value of 1.0 would triple it. We tested padding values of 0.25, 0.5, and 0.75.

3.2 Model Architectures

Three distinct CNN architectures were evaluated to provide a broad comparison:

- VGG16: A deep CNN characterised by its simple and uniform architecture, relying on stacked 3x3 convolutional layers. It serves as a powerful baseline for feature extraction. Simonyan and Zisserman, 2014
- ResNet50: A 50-layer Residual Network that introduces 'skip connections.' These connections mitigate the vanishing gradient problem and enable the effective training of much deeper networks. K. He and Sun, 2016
- EfficientNet-B0: The baseline model of the EfficientNet family, designed to achieve high accuracy and efficiency by systematically scaling model depth, width, and resolution. Tan and Le, 2019

3.3 Experimental Design and Training

The experimental pipeline was designed to isolate the effect of alignment.

- Implementation: Dlib was chosen for its speed, reliability, and robust 68-point landmark detector. The models were selected to compare a broad range of architectures.
- Input Size: All images (raw and aligned) were resized to 224x224 pixels, matching the standard input size for the chosen pre-trained models.
- Failure Handling: In cases where Dlib failed to detect a face for alignment, that image was removed from the aligned datasets. However, it was retained in the "raw" (control) dataset to ensure the control group represented the complete, unaltered data.
- Control Group: The control group used the raw, untouched dataset. For both aligned and raw datasets, model-specific normalisation transforms were applied as images were loaded.
- Training: Transfer learning was used for all models. All hidden layers were frozen, and the final output layer was replaced with a task-specific layer to detect the 40 binary facial attributes. Models were trained using PyTorch's Mean Square Error

(MSE) loss function with the Adam optimiser. A batch size of 64, a learning rate of 0.001, and a training duration of 5 epochs were used for all experiments.

4 Experimental Setup

4.1 Dataset

The CelebA dataset Z. Liu and Tang, 2015 was used for this study. It contains over 200,000 celebrity images, each annotated with 40 binary facial attributes (where a value of 1 indicates the attribute is present and -1 indicates it is not). This dataset is a standard benchmark for facial analysis, as it contains 'in-the-wild' images with moderate pose, lighting, and expression variation.

The default data partition (`list_eval_partition.txt`) was used, resulting in an 80/10/10 split for training, validation, and testing, respectively.

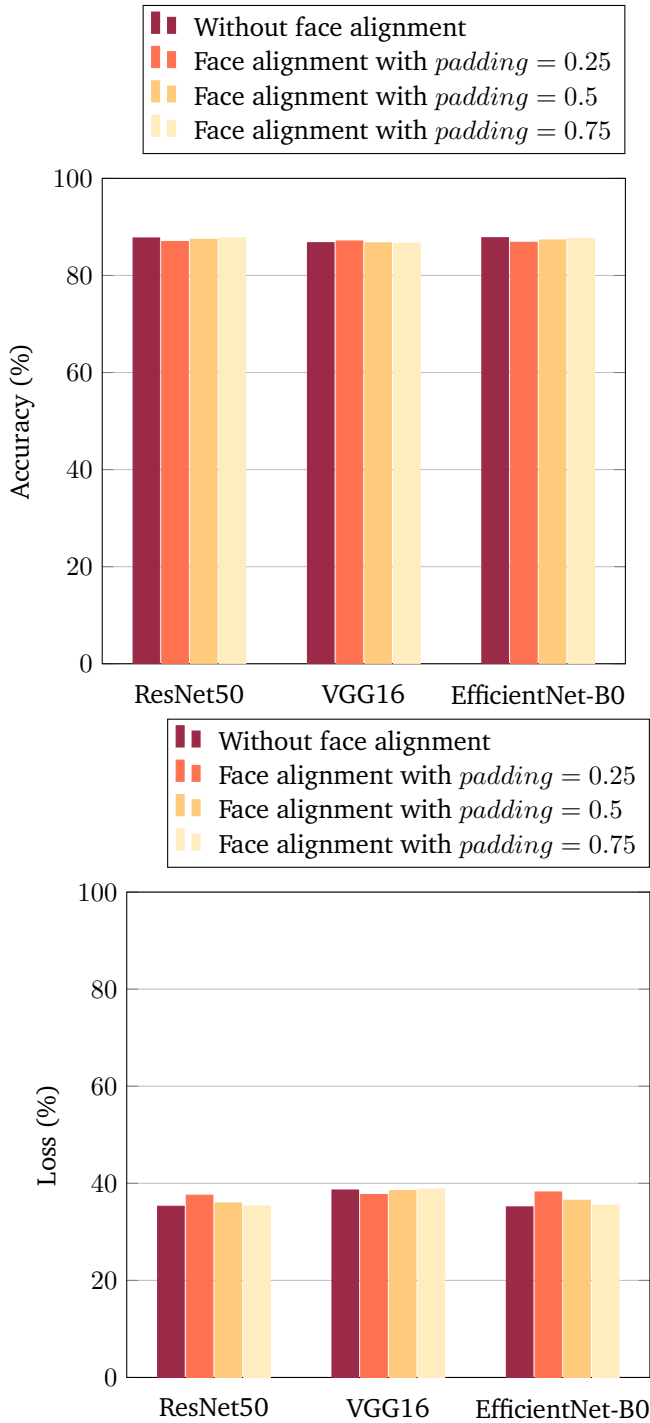
4.2 Evaluation Metrics

Performance was evaluated using two primary metrics on the test set:

- Test-Set Mean Square Error (MSE) Loss
- Test-Set Average Accuracy: which is calculated by averaging the number of correct labels predicted divided by the total number of labels.

Validation loss and accuracy were also recorded but are included in the appendices as they did not provide additional insight beyond the test-set analysis. Other metrics, such as precision, recall, and F1-score, were omitted due to the negligible difference in cost between false positives and false negatives for this particular attribute classification task.

5 Results & Discussion



As illustrated in the bar charts, the introduction of facial alignment preprocessing resulted in a minimal change in performance across all models tested.

The ResNet50 model was the only one to record any improvement, showing a negligible accuracy increase of 0.02% (from 87.72% on raw images to 87.74% with 0.75 padding). Both VGG16 and EfficientNet-B0 were observed to have slight decreases in accuracy, with a maximum drop of 0.1% and 0.94%, respectively.

Despite the insignificant overall improvement, the study's second aim revealed an interesting correlation between performance and the padding parameter.

- For ResNet50 and EfficientNet-B0, there was a positive linear correlation between the padding value and accuracy (i.e., they performed better with more focalised crops).
- Conversely, VGG16 showed a negative linear correlation, performing better with less padding (more of the surrounding image).

This finding suggests a potential locality preference for facial attributes within different model architectures. Since all models were pre-trained on ImageNet, these preferences likely stem from their inherent architectures. The newer, more complex ResNet50 and EfficientNet-B0 models may be better at generalisation and prefer a more focalised view of the features, whereas the simpler VGG16 architecture may benefit from the wider context provided by less padding.

5.1 Limitations

The limited performance boost is likely attributable to the 'clean' nature of the CelebA dataset. Most images are already relatively front-on with good lighting, making a corrective step like facial alignment largely unnecessary. The study's findings are therefore limited to this dataset, and further work is needed to see if alignment provides more significant gains on datasets with more extreme angles, occlusions, and lighting challenges.

Furthermore, the alignment process itself adds computational overhead. The cost of detecting faces and landmarks must be weighed against any potential performance gains, especially if high accuracy is not the only deployment constraint.

6 Conclusion

This report investigated the efficacy of facial alignment preprocessing by comparing the performance of ResNet50, VGG16, and EfficientNet-B0 on the CelebA facial attribute classification task.

The primary finding is that facial alignment had an insignificant impact on the performance of all tested network architectures, resulting in a slight average accuracy decrease of 0.08.

However, a secondary finding revealed positive linear correlations between accuracy and increased padding for the ResNet50 and EfficientNet-B0 models. This suggests that while domain-specific preprocessing like alignment may have a limited impact on high-quality datasets, the way features are presented (i.e., focalised vs. contextual) does affect performance, with newer models potentially benefiting more from a tightly-cropped, focalised area of interest.

Future work should validate these findings on more challenging "in-the-wild" datasets to determine if the benefits of facial alignment become more pronounced as image quality and pose variation decrease.

7 Appendix

7.1 Appendix A: Testing Results

Table 1: Testing Accuracy

	ResNet50	VGG16	EfficientNet-B0
Without	87.72%	86.74%	87.77%
Padd.=0.25	86.99%	87.10%	86.83%
Padd.=0.5	87.45%	86.72%	87.31%
Padd.=0.75	87.74%	86.64%	87.61%

Table 2: Testing Loss

	ResNet50	VGG16	EfficientNet-B0
Without	35.27%	38.62%	35.14%
Padd.=0.25	37.56%	37.69%	38.21%
Padd.=0.5	35.95%	38.50%	36.51%
Padd.=0.75	35.32%	38.81%	35.50%

7.2 Appendix B: Validation results without facial alignment

Table 3: Validation Accuracy Without Face Alignment

	ResNet50	VGG16	EfficientNet-B0
Epoch 1	88.22%	86.35%	88.06%
Epoch 2	88.12%	87.08%	88.28%
Epoch 3	88.19%	87.22%	88.29%
Epoch 4	88.19%	87.49%	88.31%
Epoch 5	88.24%	87.27%	88.31%

Table 4: Validation Loss Without Face Alignment

	ResNet50	VGG16	EfficientNet-B0
Epoch 1	33.96%	40.06%	34.54%
Epoch 2	34.20%	37.46%	33.75%
Epoch 3	34.05%	37.07%	33.88%
Epoch 4	34.13%	36.80%	33.81%
Epoch 5	34.05%	37.29%	33.87%

7.3 Appendix C: Validation results with facial alignment and padding of 0.25

Table 5: Validation Accuracy With Facial Alignment and Padding of 0.25

	ResNet50	VGG16	EfficientNet-B0
Epoch 1	87.53%	86.34%	87.27%
Epoch 2	87.53%	87.56%	87.45%
Epoch 3	87.62%	87.49%	87.51%
Epoch 4	87.54%	87.61%	87.48%
Epoch 5	87.56%	87.75%	87.48%

Table 6: Validation Loss With Facial Alignment and Padding of 0.25

	ResNet50	VGG16	EfficientNet-B0
Epoch 1	36.25%	36.69%	37.24%
Epoch 2	36.15%	36.41%	36.53%
Epoch 3	35.97%	36.29%	36.43%
Epoch 4	36.20%	36.08%	36.43%
Epoch 5	36.14%	35.98%	36.55%

7.4 Appendix D: Validation results with facial alignment and padding of 0.5

Table 7: Validation Accuracy With Facial Alignment and Padding of 0.5

	ResNet50	VGG16	EfficientNet-B0
Epoch 1	87.98%	86.27%	87.59%
Epoch 2	87.92%	87.01%	87.91%
Epoch 3	87.93%	87.41%	87.78%
Epoch 4	87.96%	87.31%	87.85%
Epoch 5	87.83%	87.27%	87.93%

Table 8: Validation Loss With Facial Alignment and Padding of 0.5

	ResNet50	VGG16	EfficientNet-B0
Epoch 1	34.82%	40.38%	36.01%
Epoch 2	34.88%	37.40%	34.97%
Epoch 3	34.72%	36.62%	35.37%
Epoch 4	34.82%	37.38%	35.13%
Epoch 5	35.10%	37.22%	35.07%

7.5 Appendix E: Validation results with facial alignment and padding of 0.75

Table 9: Validation Accuracy With Facial Alignment and Padding of 0.75

	ResNet50	VGG16	EfficientNet-B0
Epoch 1	88.07%	86.25%	87.89%
Epoch 2	88.28%	86.89%	88.18%
Epoch 3	88.29%	87.19%	88.18%
Epoch 4	88.22%	87.15%	88.15%
Epoch 5	88.22%	87.03%	88.14%

Table 10: Validation Loss With Facial Alignment and Padding of 0.75

	ResNet50	VGG16	EfficientNet-B0
Epoch 1	34.28%	40.64%	34.92%
Epoch 2	33.85%	37.72%	34.16%
Epoch 3	33.79%	37.37%	34.23%
Epoch 4	33.87%	37.42%	34.14%
Epoch 5	33.40%	37.74%	34.23%

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